kickstart_lecture4_matplotlib

December 19, 2024

0.1 Python Kickstart Lecture 4: Matplotlib for Plotting

[1]:
```
# Import libraries: Numpy and Matplotlib

import numpy as np
import matplotlib.pyplot as plt
```

```
Intel MKL WARNING: Support of Intel(R) Streaming SIMD Extensions 4.2 (Intel(R)
SSE4.2) enabled only processors has been deprecated. Intel oneAPI Math Kernel
Library 2025.0 will require Intel(R) Advanced Vector Extensions (Intel(R) AVX)
instructions.
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instructions.
```
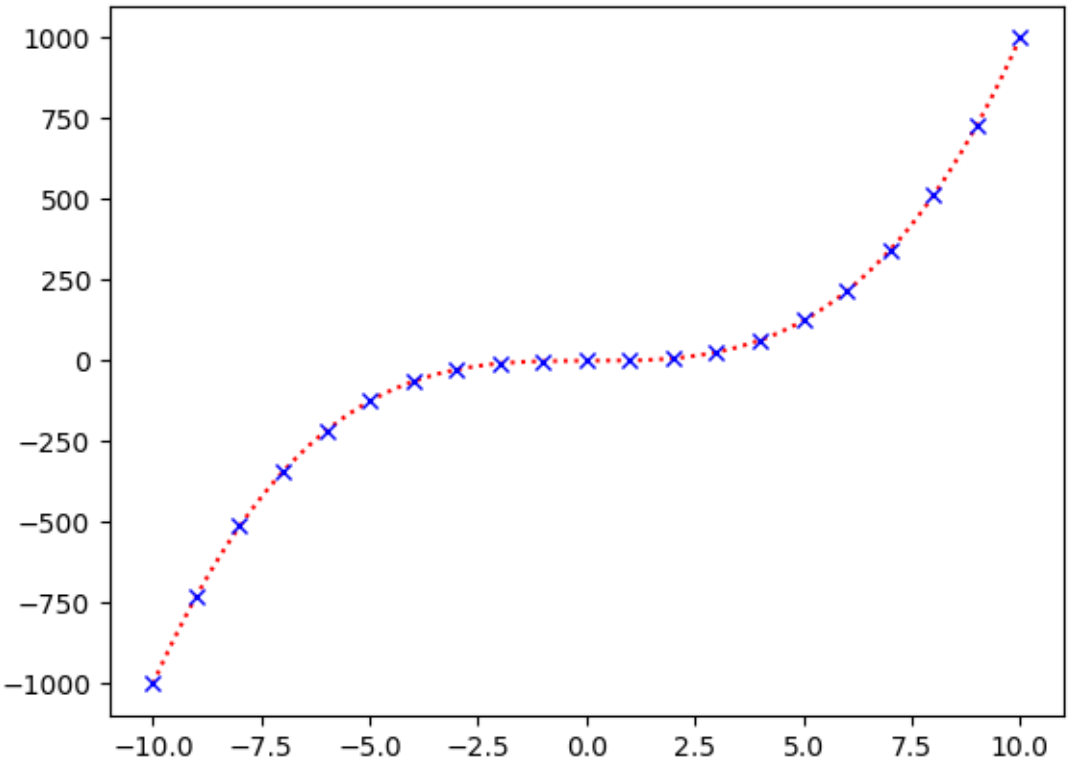

0.1.1 Line Plot

[2]:
```
# Generate random x-y Data

x = np.linspace(-10,10,201)
y = x**3-1
```

[3]:
```
x2 = np.linspace(-10,10,21)
y2 = x2**3-1
```

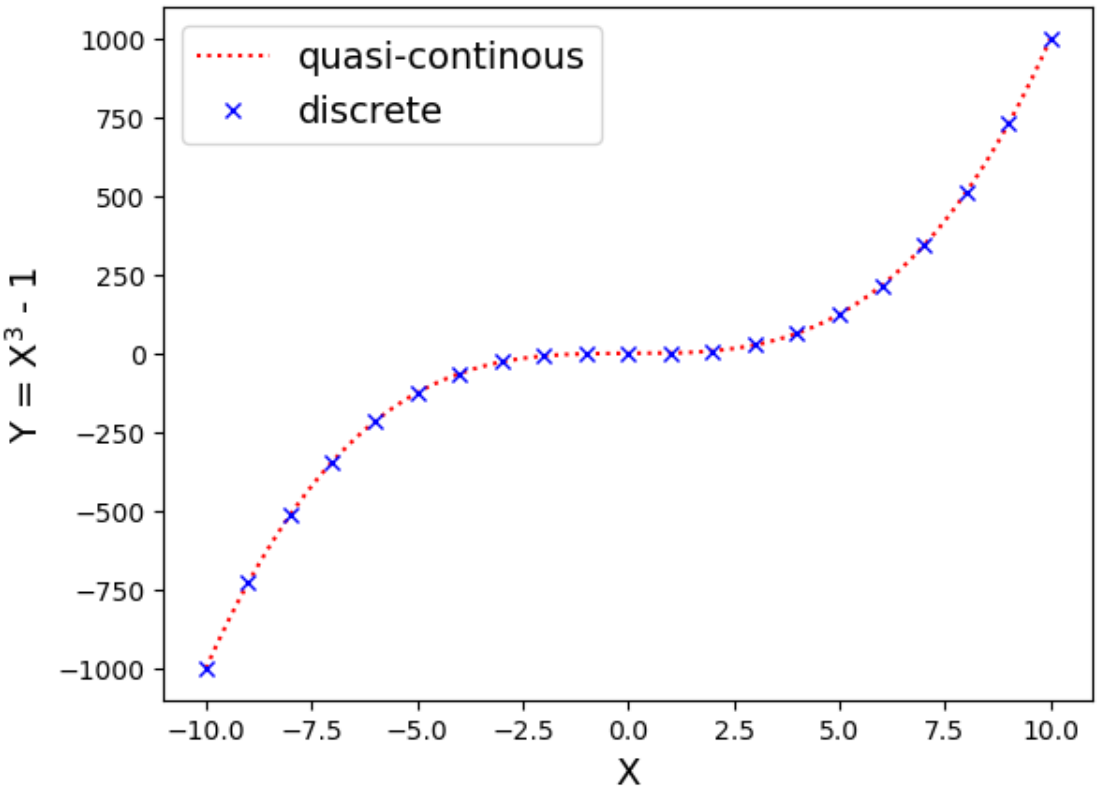
[9]:
```
# Create plot

plt.figure()
plt.plot(x,y, ':', color = 'red')
plt.plot(x2,y2, 'x',color = 'blue')
```

[9]:
```
[<matplotlib.lines.Line2D at 0x11700ee70>]
```



[14]:
```
# Add axis labels, legends, fontsize

plt.figure()
plt.plot(x,y, 'r:', label = 'quasi-continous')
plt.plot(x2,y2, 'bx',label = 'discrete')
plt.legend(fontsize = 14)
plt.xlabel('X', fontsize = 14)
plt.ylabel('Y = X$^3$ - 1', fontsize = 14)
```

[14]:
```
Text(0, 0.5, 'Y = X$^3$ - 1')
```



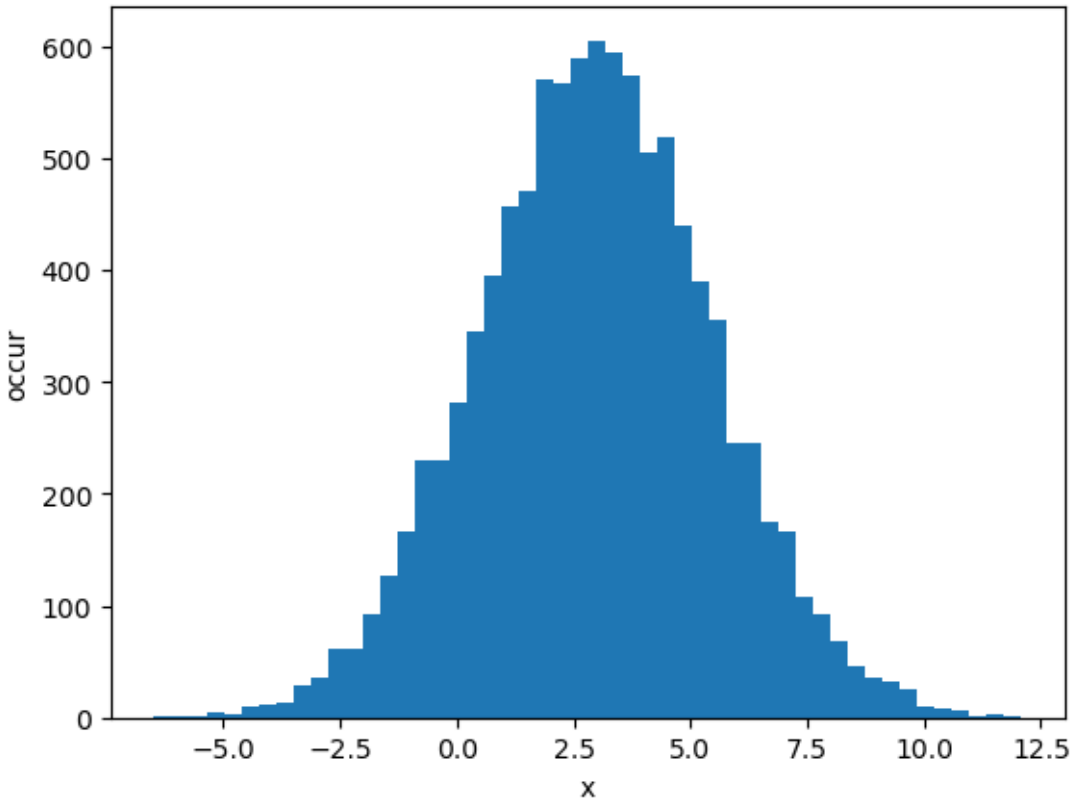
[ ]:
```
# Adjust color, linestyle -> see above
```

0.1.2 Histogram

[26]:
```
# Generate gaussian distributed data

gaussian_data = 3 + 2.5 * np.random.randn(10000)
```

[27]:
```
plt.figure()
plt.hist(gaussian_data, 50)
plt.xlabel('x')
plt.ylabel('occur')
```

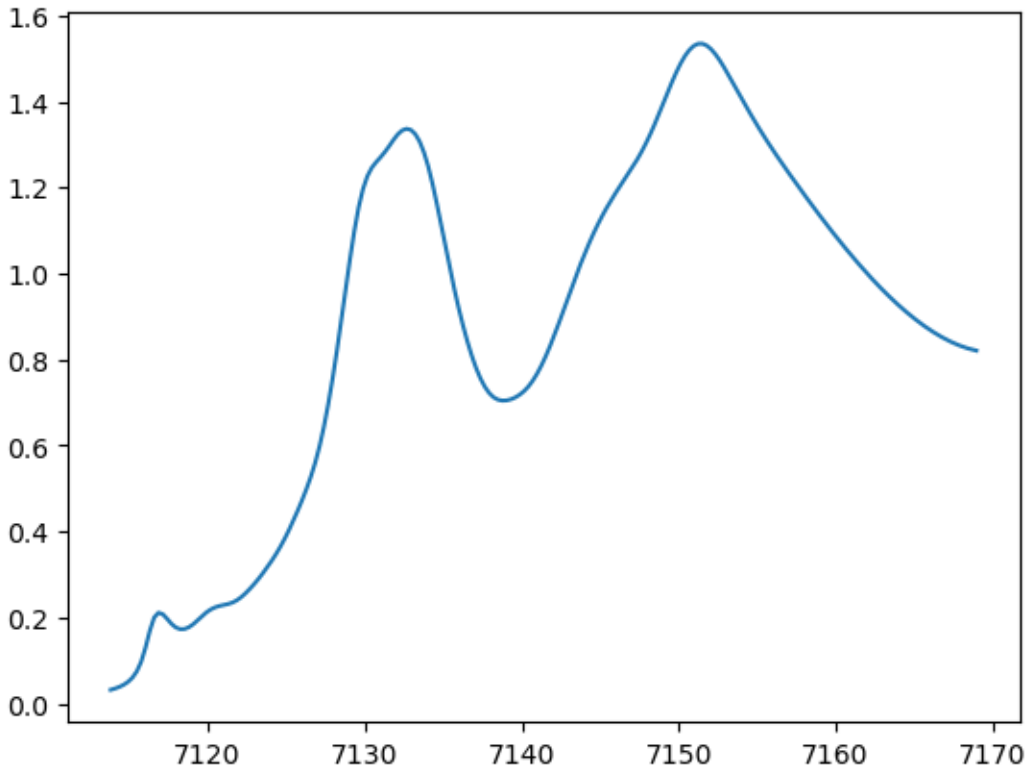
[27]:
```
Text(0, 0.5, 'occur')
```



0.1.3 Loading Data and Visualizing

[36]:
```
# Load XAS data with loadtxt

data = np.loadtxt('xas_FeCO5_from_materialsproject.tsv',
          delimiter='\t',
          skiprows = 4,
          usecols = [0,1]
          )
x_data = data[:,0]
y_data = data[:,1]
```

[39]:
```
# Create a plot
plt.figure()
plt.plot(x_data,y_data)

plt.savefig('Fe_xas.png', dpi = 300) # pixelated
plt.savefig('Fe_xas.eps') # vectorized
```



[38]:
```
# And save the plot to png and eps
```

```
<Figure size 640x480 with 0 Axes>
```

[ ]:
```

```